MUNISOLVE ZA

Municipal faults, reported and resolved in public

YEAR	ROLE	TEAM	STATUS
2026	Full-stack engineering, product design	Personal project	Live

Summary

A municipal service-delivery platform where citizens report faults, administrators triage them on a live heat map and an assistant answers common questions within a rate limit.

The problem

A broken streetlight is easy to report and hard to track. Municipal teams need a shared picture of where faults are, and residents need to know that a report went somewhere.

MuniSolve ZA gives citizens a simple reporting flow and administrators a live operational view. The assistant, Siyanda, handles the questions that do not need a person, with a visible limit so that it cannot become an unbounded cost centre.

What is built

- JWT, bcrypt and Google OAuth, Helmet and rate limiting, and role-based access (citizen, admin, super-admin) with activity logs.
- A Leaflet heat map plus Open-Meteo and Nager.Date widgets; deployed on Vercel, Render and Neon.

Decisions that matter

Three roles, logged actions

Choice. Citizen, admin and super-admin permissions are enforced server-side with activity logs.

Why. Every consequential action has an owner.

Trade-off. More permission paths to test.

A bounded assistant

Choice. Siyanda is limited to thirty messages per hour.

Why. The assistant cannot become an unbounded cost centre.

Trade-off. Some questions wait or go to a person.

One shared operational picture

Choice. Faults appear on a live map for administrators.

Why. Teams triage from the same view.

Trade-off. Location data needs care.

Stack

`React 19` `Node.js` `Express` `PostgreSQL` `Prisma` `Claude`

Where to look

- Open live site: https://munisolve-za.vercel.app
- Source on GitHub: https://github.com/konethegreat/munisolve-za
- Case study: konethegreat.github.io/work/munisolve